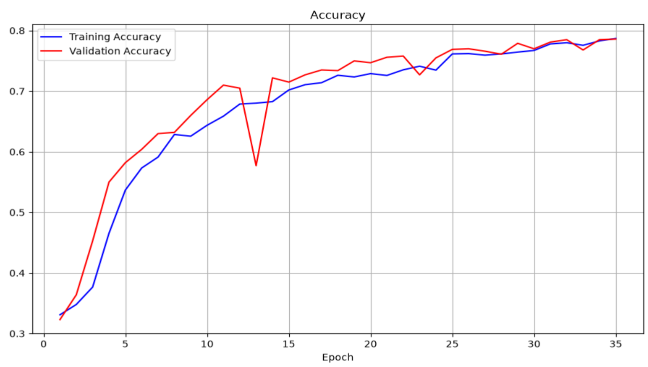
P6 Sections 5 & 6 — Facial Recognition CNN

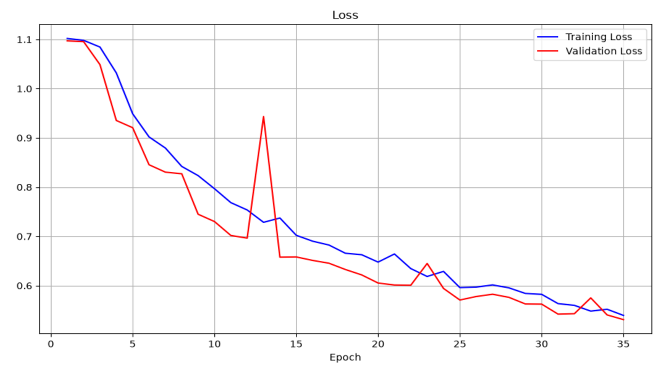
Dataset: facial expression images (happy / neutral / surprise), 5000 training images via export_dataset.py.

Network Summary

Rescaling → RandomFlip → RandomRotation → Conv(24)-Pool → Conv(32)-Pool → Conv(48)-Pool → Conv(64)-Pool → Dropout(0.25) → Flatten → Dense(32) → Dropout(0.4) → Softmax(3). Optimizer: Adam (lr=0.0009).
Total parameters: **149,683** (limit 150,000).
Epochs trained: **35** (EarlyStopping + ReduceLROnPlateau).
Best validation accuracy: **0.7860**.
Held-out test accuracy: **0.7775**, test loss: **0.5581**.
Saved model: **results/best_basic_model.keras**.

Training / Validation Curves





Hyperparameter Optimization Strategy (Section 6)

- Swept convolutional widths (16–64) while staying under 150k parameters.
- Compared Dense sizes 16 vs 32; Dense(32) improved capacity without exceeding the limit.
- Inserted dropout after the last conv stack (0.25) and after Dense (0.4).
- Added light augmentation (horizontal flip + small rotation).
- Switched optimizer to Adam (lr=0.0009) with ReduceLROnPlateau.
- Used EarlyStopping on val_accuracy to keep the best epoch before overfitting.

Example Training Images

neutral

happy

surprise